SOFTWARE REQUIREMENTS SPECIFICATION

Detailed SRS — IEEE-style structure adapted for the project

AI-Powered Multimodal Assistive Application for People with Mild Cognitive Impairment (MCI)

Mridula J B, Navaneeth S Narayan, Neeraj Ram B, Sonal Santhosh

Prof. Binu Rajan M R, Assistant Professor, Dept. of CSE, SCTCE

Revision History

Version	Date	Description
1.0	24 September 2026	Initial detailed SRS baseline

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification is to define the behavior and quality requirements of the AI-powered multimodal assistive application before implementation. It provides a common reference for the development team, guide, testers and evaluators.

1.2 Product Scope

The application combines computer vision, conversational memory retrieval, scheduling/reminders, safety monitoring and caregiver support in one system. The supplied project presentation describes the intended architecture as a mobile capture layer, Django backend, independent AI modules and a decision engine feeding patient and caregiver interfaces.

1.3 Intended Audience

- Project development team
- Project guide and review committee
- Software testers
- Future maintainers
- Academic evaluators

1.4 Definitions

Term	Definition
MCI	Mild Cognitive Impairment.
RAG	Retrieval-Augmented Generation: retrieval of relevant stored information before generating a response.
Embedding	Numerical vector representation used for similarity search.
Geofence	Virtual geographic boundary used to detect entry/exit conditions.
Last-seen location	Most recent location/context at which an object was successfully detected.
Familiar person	A person registered by/for the patient with identity and relationship information.
Caregiver	Authorized person responsible for configuring and monitoring patient assistance.

2. Overall Description

2.1 Product Perspective

The product is a multimodal assistive application rather than a standalone chatbot. It accepts camera, audio and location information, processes those inputs through specialized modules, stores relevant structured/unstructured data, and provides context-aware assistance.

2.2 Product Functions

- Authentication and role-based access.
- Patient profile and caregiver linkage.
- Object detection, tracking and object-memory retrieval.
- Familiar-person face recognition and relationship cues.
- Conversation capture, speech-to-text, memory indexing and RAG retrieval.
- Personal AI assistant.
- Calendar, daily routine, appointments and reminders.
- Medication schedules and reminders.
- GPS monitoring and geofencing.
- Caregiver dashboard and alerts.
- Relaxation/meditation/well-being support.

2.3 User Characteristics

User	Characteristics / Needs
Patient	Requires simple, low-cognitive-load interaction, clear feedback, optional voice assistance and minimal navigation.
Caregiver	Needs more detailed configuration, monitoring, notifications and management controls.

2.4 Operating Environment

- Android/iOS smartphone or supported mobile device for patient-side capture and interaction.
- Backend server/local processing node for application APIs and AI services.
- Camera and microphone for multimodal capture.
- Location services/GPS for geofencing.
- Database and vector storage for structured records and memory retrieval.

3. External Interface Requirements

3.1 Mobile UI

- Patient home/dashboard.
- Schedule/calendar view.
- Reminder/medication view.
- Object finder.
- People/familiar-person view.
- AI assistant/chat/voice interface.
- Relaxation/well-being section.
- Safety/status section.

3.2 Caregiver UI

- Patient overview.
- Schedule and appointment management.
- Medication management.
- Familiar-person registration.
- Geofence configuration.
- Alerts and notifications.

- Activity history.

3.3 Hardware Interfaces

Interface	Requirement
Camera	Capture images/video frames for object and face processing.
Microphone	Capture supported voice input for speech recognition.
GPS/Location	Provide location information for geofencing and location-aware features.
Speaker	Provide voice responses, reminders and relaxation audio where supported.

3.4 Software Interfaces

Component	Role
Mobile application	User interface and multimodal capture.
Django/REST backend	Authentication, business logic, APIs and orchestration.
Computer vision services	Object and face processing.
Speech-to-text	Convert supported voice input to text.
Embedding model	Generate vectors for memory retrieval.
Vector database/index	Similarity search over stored memories.
LLM	Generate contextual responses from retrieved/structured information.
Notification service	Deliver reminders and caregiver alerts.

4. Detailed Functional Requirements

4.1 Authentication and Authorization

AUTH-01: System shall allow patient/caregiver authentication.

AUTH-02: System shall enforce role-based access.

AUTH-03: System shall terminate/expire invalid sessions according to configured security policy.

AUTH-04: Caregiver shall only access patients for whom permission exists.

4.2 Patient Profile and Familiar People

PEO-01: Caregiver shall create a familiar-person profile.

PEO-02: Profile shall include name and relationship.

PEO-03: System shall associate supported face-recognition data with the profile.

PEO-04: System shall allow profile update/revocation.

4.3 Object Memory

OBJ-01: System shall receive supported camera frames.

OBJ-02: System shall detect configured everyday objects.

OBJ-03: System shall track object observations across supported frames.

OBJ-04: System shall associate an observation with timestamp and approximate scene/location context.

OBJ-05: System shall maintain the latest valid observation.

OBJ-06: Patient shall be able to query an object by name or voice.

OBJ-07: System shall return the last detected location and timestamp.

OBJ-08: System shall state/indicate when no recent observation is available.

4.4 Facial Recognition

FACE-01: System shall detect faces from supported camera input.

FACE-02: System shall generate a recognition representation for a detected face.

FACE-03: System shall compare the representation against registered familiar people.

FACE-04: System shall display identity and relationship only when confidence is sufficient.

FACE-05: System shall treat insufficient-confidence matches as unknown.

4.5 Conversational Memory and RAG

RAG-01: System shall accept typed or supported voice queries.

RAG-02: System shall convert supported voice queries to text.

RAG-03: System shall identify relevant context such as person, time and topic where available.

RAG-04: System shall retrieve semantically relevant stored memories.

RAG-05: System shall use recognized-person context as an optional retrieval filter/weight.

RAG-06: System shall generate a response using retrieved memories and structured application data.

RAG-07: System shall avoid presenting unavailable information as a known fact.

4.6 Schedule, Calendar and Reminders

SCH-01: Caregiver shall create recurring or one-time schedule items.

SCH-02: Patient shall view the current day's schedule.

SCH-03: Caregiver shall create appointments.

SCH-04: System shall generate reminders at configured times.

SCH-05: Patient shall be able to mark/confirm supported reminders as completed.

SCH-06: Caregiver shall be able to review reminder status where implemented.

4.7 Medication

MED-01: Caregiver shall create a medication record.

MED-02: Medication record shall include name, dosage/instructions and schedule.

MED-03: System shall generate medication reminders.

MED-04: System shall record reminder status/confirmation where supported.

MED-05: AI shall not change medication dosage or prescribe medication.

4.8 Geofencing and Safety

SAFE-01: Caregiver shall define a safe zone.

SAFE-02: System shall obtain supported location updates.

SAFE-03: System shall determine whether a location is inside/outside an active safe zone.

SAFE-04: System shall generate an alert on a configured violation.

SAFE-05: Alert shall include relevant time and location information.

4.9 Well-being Support

WELL-01: Patient shall be able to browse supported relaxation/meditation content.

WELL-02: Patient shall be able to start/stop a session.

WELL-03: System may use contextual cues to suggest well-being content, but suggestions shall not be presented as medical treatment.

4.10 Caregiver Dashboard

CARE-01: Caregiver shall view patient status.

CARE-02: Caregiver shall view schedule and reminder information.

CARE-03: Caregiver shall receive configured alerts.

CARE-04: Caregiver shall manage geofences.

CARE-05: Caregiver shall manage familiar-person profiles.

CARE-06: Caregiver shall review relevant activity history.

5. Non-Functional Requirements

5.1 Security

The system shall protect authentication credentials, restrict data by role, secure API communication and protect sensitive patient data.

5.2 Privacy

Raw audio/video and location data shall not be retained longer than required by the defined feature. Sensitive data shall be accessible only to authorized users.

5.3 Usability

The patient interface shall minimize cognitive load through simple flows, large controls, clear language, consistent icons and optional voice interaction.

5.4 Performance

AI pipelines shall be benchmarked for latency on the target hardware. Performance targets shall be recorded after prototype measurement rather than assumed.

5.5 Reliability

A failure in one AI component should degrade that feature gracefully rather than bring down core scheduling/reminder functions.

5.6 Maintainability

Modules shall have clear service boundaries and interfaces; model implementations should be replaceable without rewriting unrelated modules.

5.7 Safety

AI-generated information shall be grounded in stored data and shall communicate uncertainty where relevant.

6. System Behavior and Business Rules

1. BR-01: A patient may have one or more authorized caregivers according to the application's permission model.
2. BR-02: A familiar-person profile must exist before the system can reliably identify that person.
3. BR-03: An object answer refers to the latest valid observation, not a guaranteed current location.
4. BR-04: A low-confidence face match must not be presented as a confirmed identity.
5. BR-05: Structured data such as medication schedules should be retrieved directly from the database rather than invented by the LLM.
6. BR-06: RAG responses should be grounded in retrieved memories and should indicate lack of information when appropriate.
7. BR-07: Geofence alerts depend on location service availability and configured rules.
8. BR-08: AI assistance is supportive and does not replace clinical judgment or emergency services.

7. Error and Exception Handling

Scenario	Expected Behavior
Camera unavailable	Explain that camera access is unavailable and allow non-camera features to continue.
Microphone unavailable	Allow typed interaction and display a clear voice-input error.
No object detected	State that the object could not be located from available observations.
Face confidence too low	Identify as unknown / ask user to try again; do not guess.
No relevant memory	State that no relevant stored memory was found.
LLM unavailable	Fallback to structured responses for schedules/reminders where possible.
GPS unavailable	Disable/flag geofence monitoring and notify caregiver if configured.
Notification service unavailable	Persist pending notification state and retry where supported.

8. Security and Privacy Requirements

- Use authenticated sessions and role-based authorization.
- Protect face embeddings and conversation data as sensitive information.
- Use secure transport between mobile and backend.

- Store only data required for project functionality.
- Provide caregiver access only to authorized patient records.
- Record important administrative/security events.
- Do not expose internal prompts, embeddings or raw sensitive data unnecessarily.

9. Verification Strategy

Every requirement should be verified using one or more of the following:

- Inspection — documentation/UI/configuration review.
- Functional test — execute a defined input and compare output.
- Integration test — verify interaction between modules.
- Performance test — measure latency/resource usage.
- Security test — verify authorization and access boundaries.

10. Traceability

The requirement IDs in this SRS should be reused in the design document, API specification, implementation issue list and test cases. This creates a direct chain from requirement → design → code → test → demonstration.